Geocomputation

Spatial Queries and Geometric Operations



Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

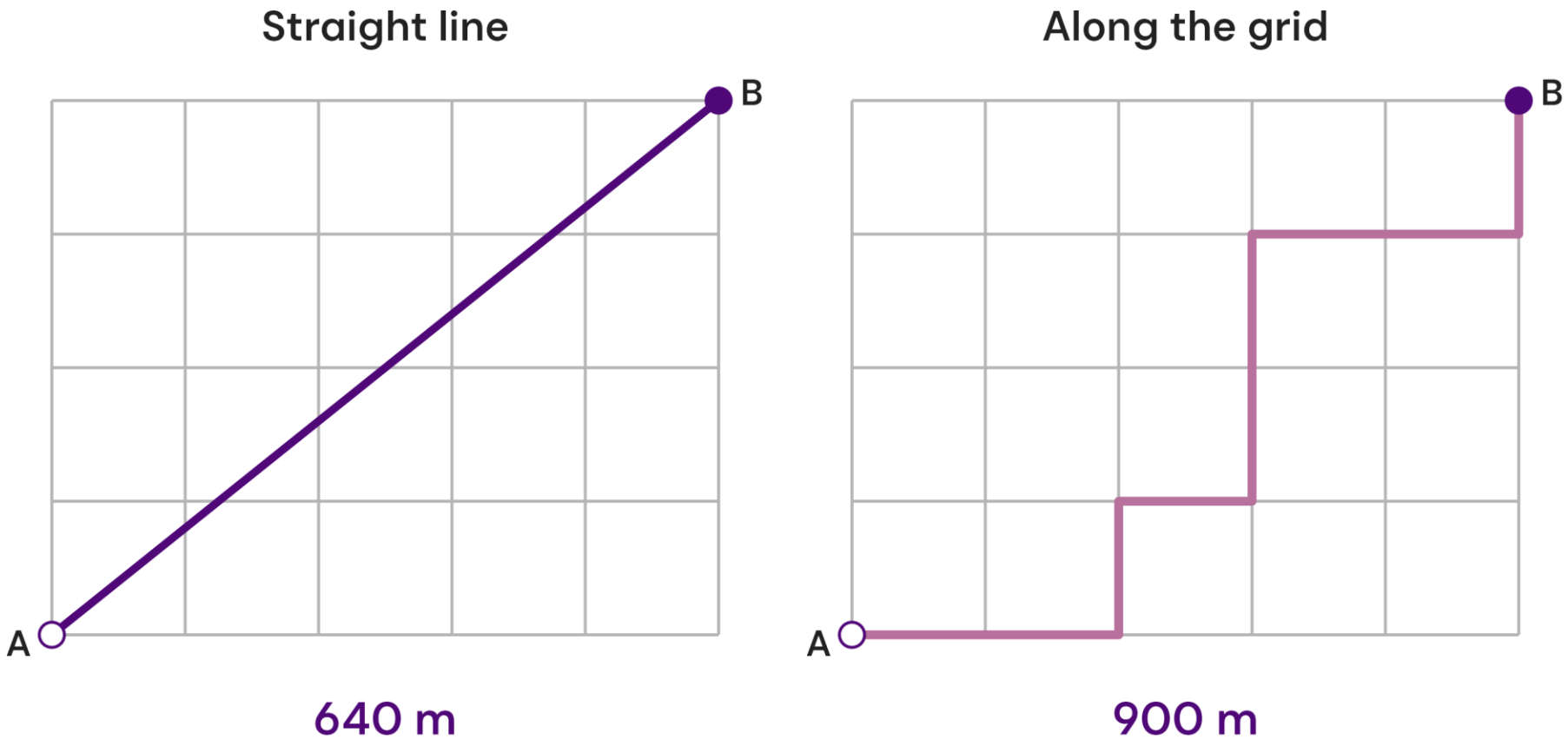
This week

- Spatial properties
- Attribute joins
- Spatial operations

Spatial properties

- Spatial analysis involves performing mathematical operations on spatial properties.
- Key properties include areas of and distances between spatial objects.
- There are diverse ways to conceptualise and work with these properties depending on the analytical context and available data.

Distance: straight line or along a grid



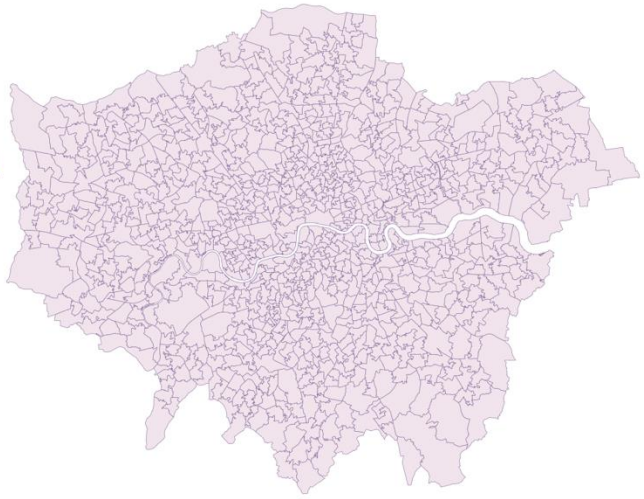
Invented example: a grid of streets, one block is 100 m.

Area

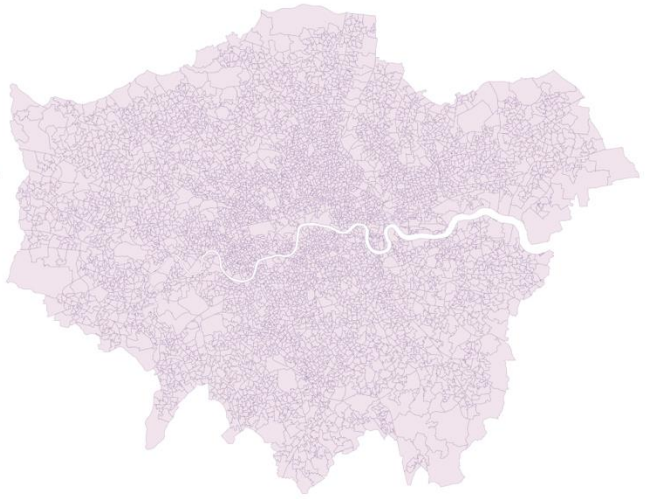
33 boroughs



1,002 MSOAs



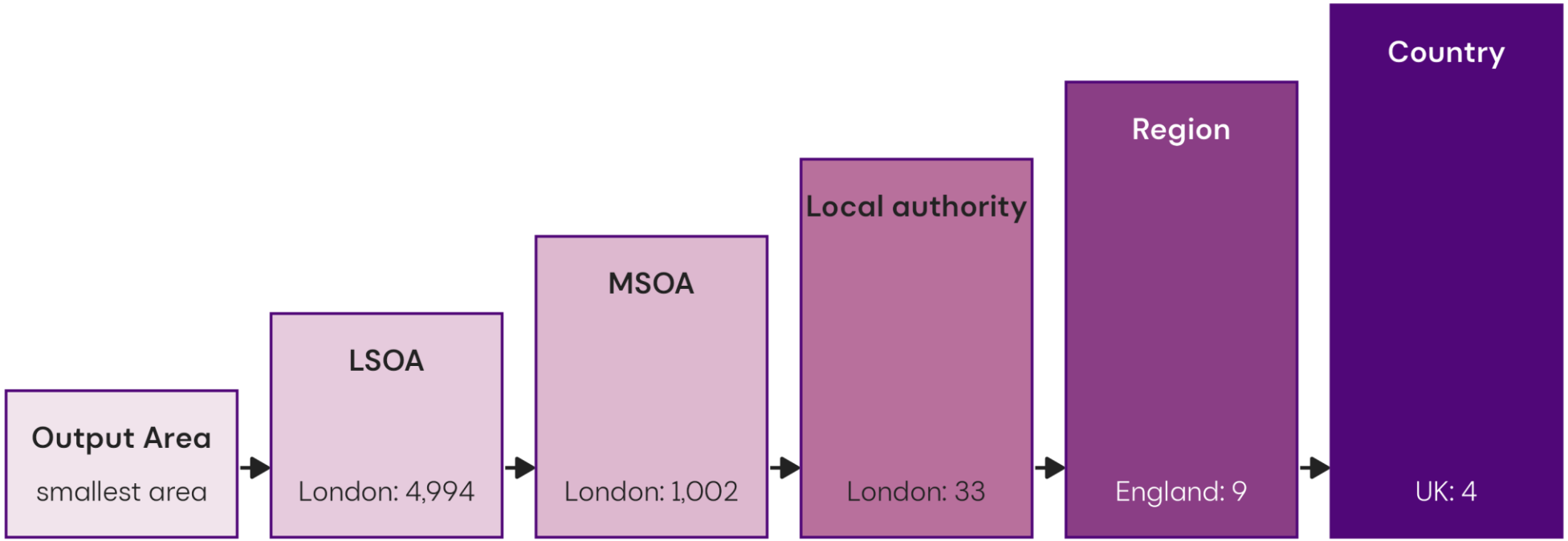
4,994 LSOAs



The same city at three levels of detail

Source: ONS boundaries for London.

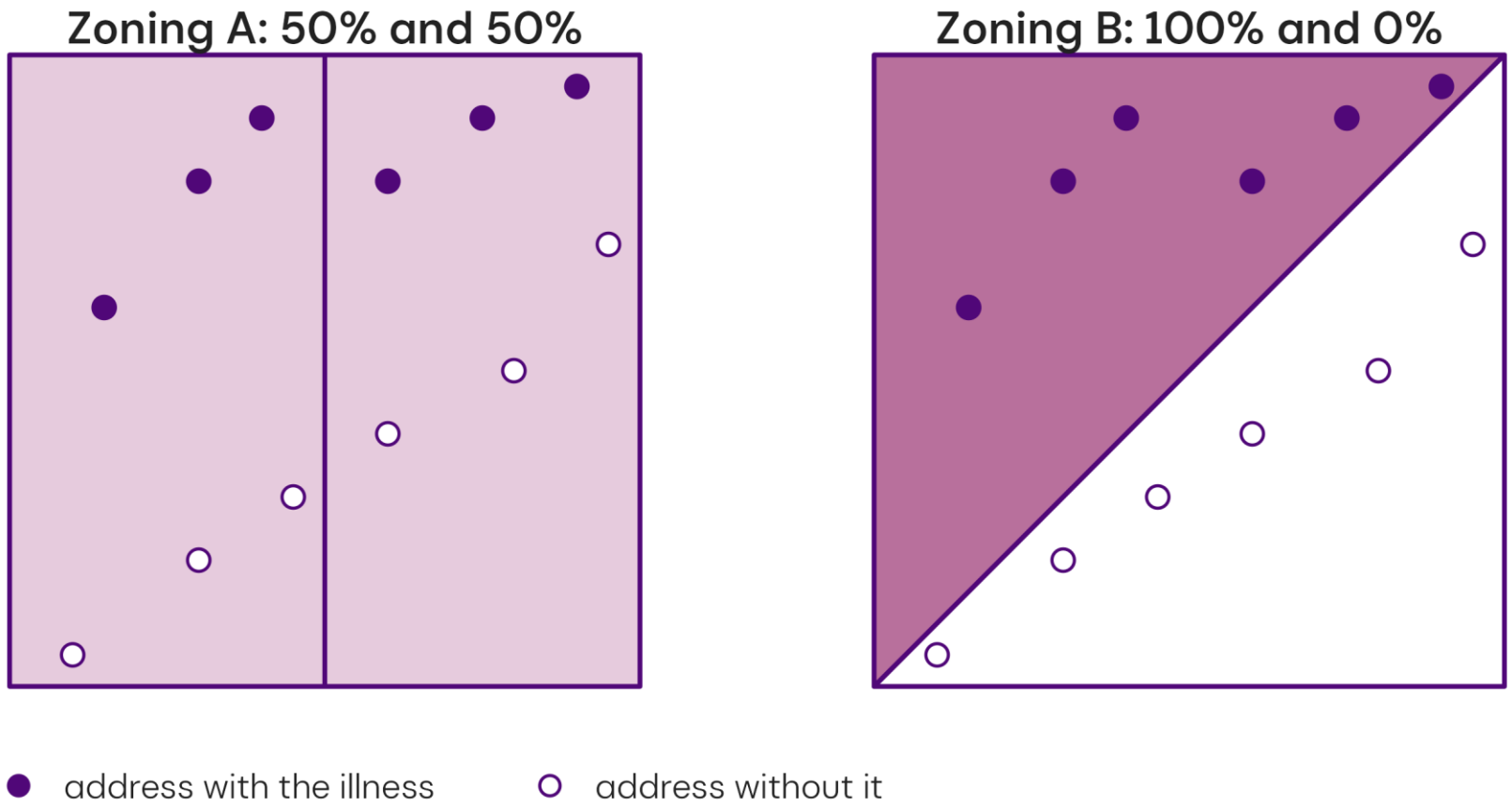
Area: UK statistical geographies



Smaller areas add up to larger ones

London counts from ONS 2021 boundaries.

Area: the modifiable areal unit problem



Invented example: 12 addresses, 6 with an illness.

Spatial features

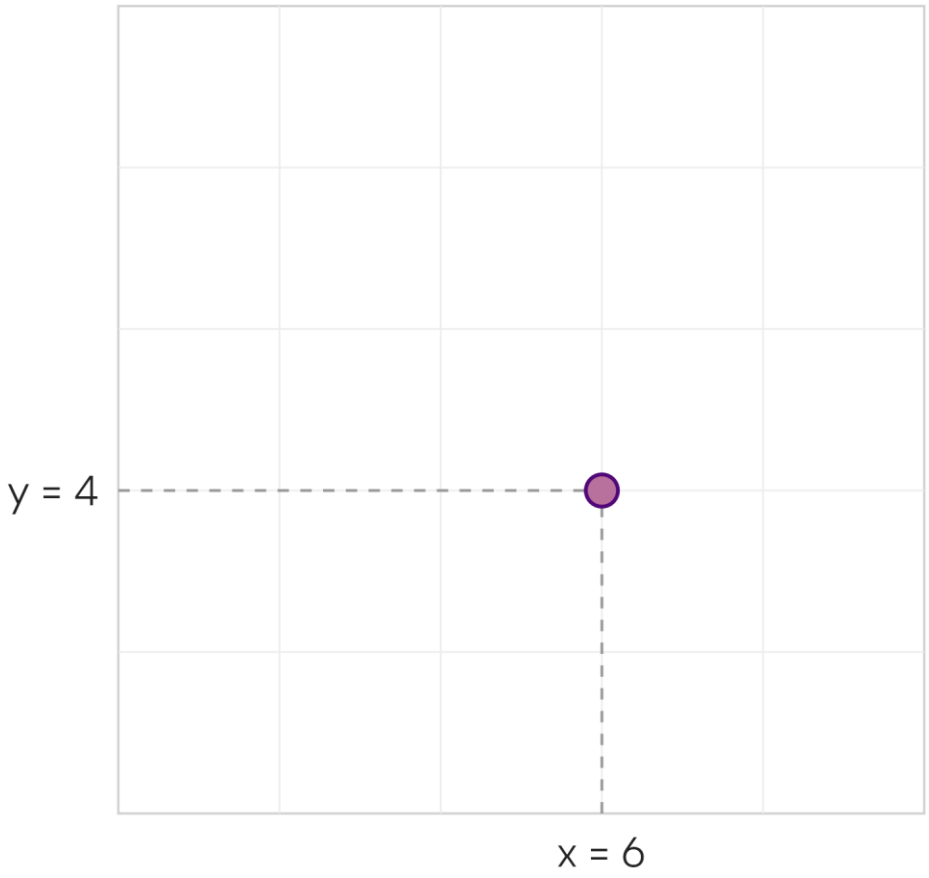
- Features: Objects created in a spatial dataset, representing geographic entities such as points, lines, or polygons.
- Attributes: Characteristics or properties of features, often stored as data in GIS, describing specific details about the feature.

Point vector

Characteristics of a point vector in a GIS data model:

- Single XY location (coordinate).
- Has no area.
- Has no length.
- Geometry consists of a single node or vertex.
- Used for: discrete features or events.

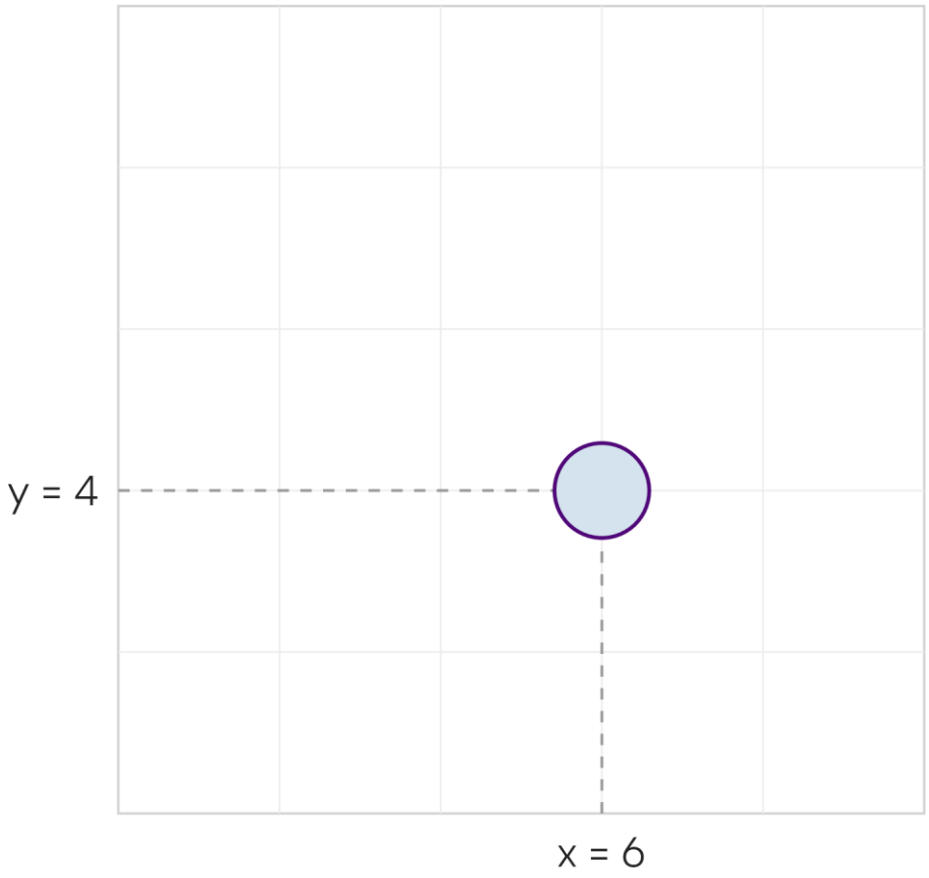
Point vector



One location

A single x and a single y.

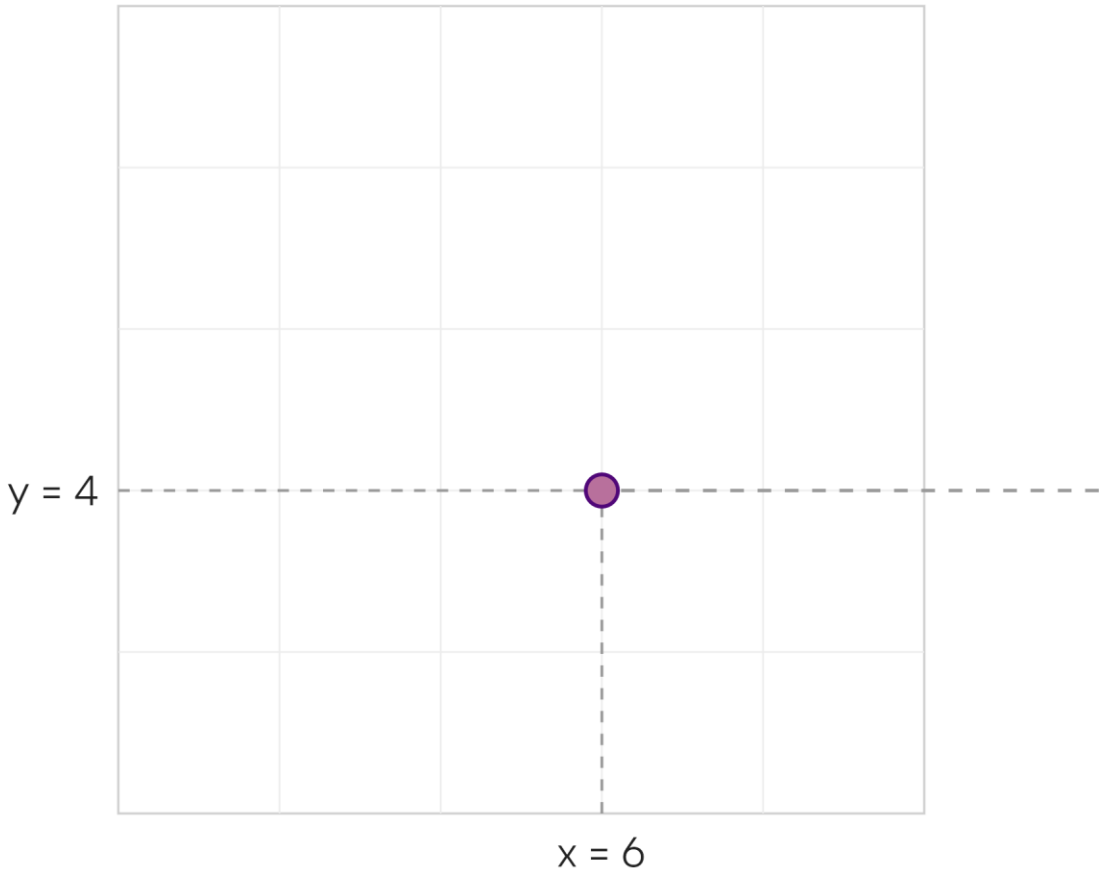
Point vector



Same location

Only the symbol changed.

Point vector



Attribute table

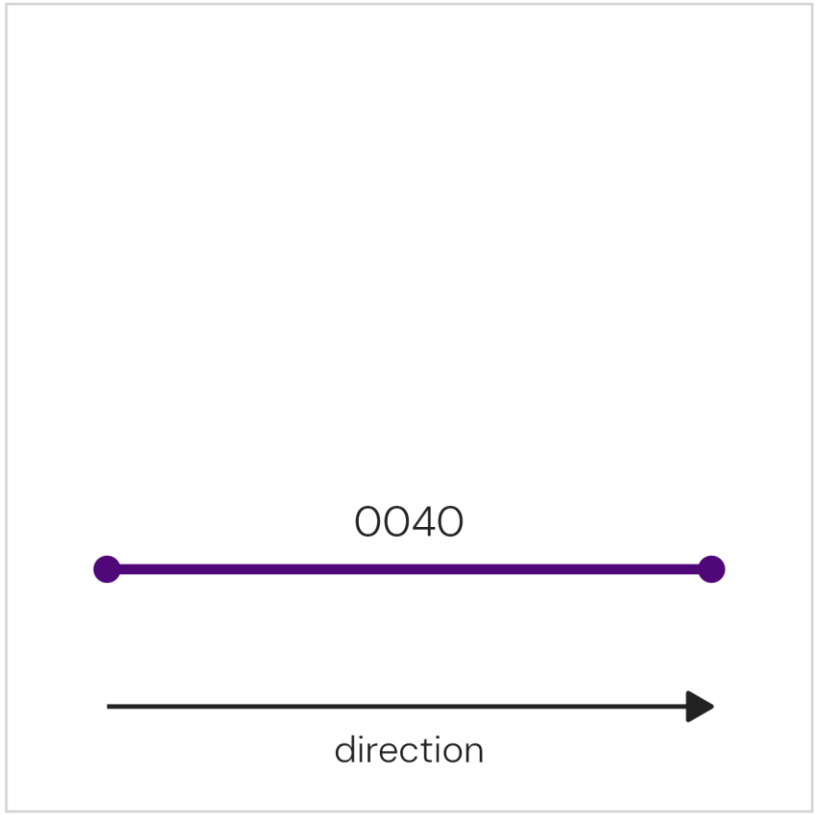
FeatureID	Type	Height
0030	Ent	500

Polyline vector

Characteristics of a polyline vector in a GIS data model:

- Series of XY locations (coordinates) that form a line.
- Has no area.
- Has a length.
- Has a direction (importance when it comes to roads, rivers, etc.).
- Can be connected to other polyline vectors to form a network.
- Geometry consists of two nodes (start node and end node) and can have one or more vertices.
- Used for: features without an area but with a length.

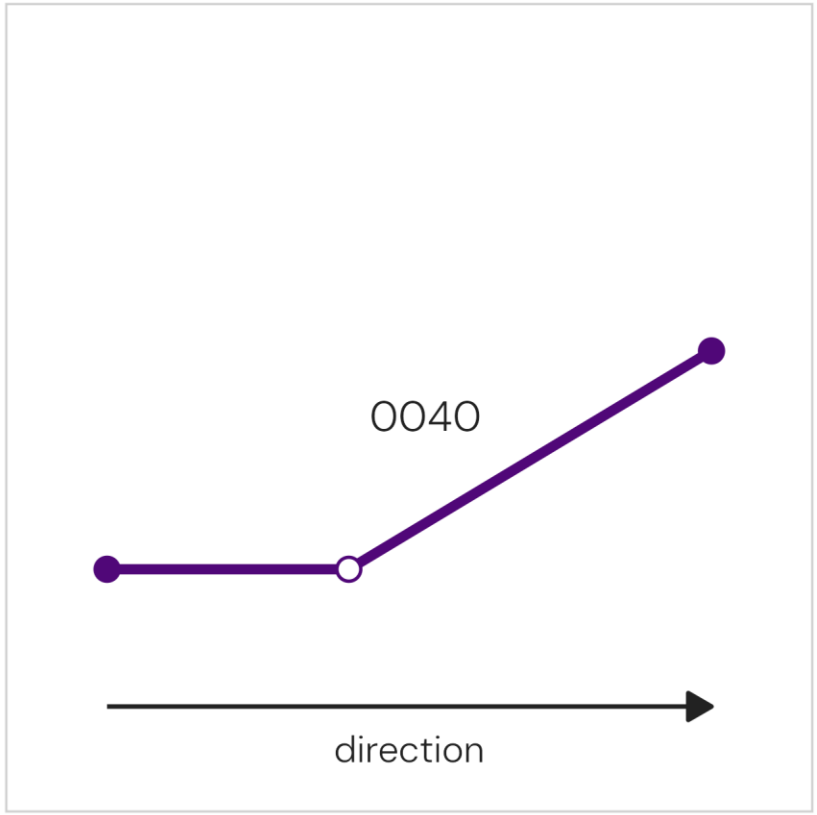
Polyline vector



FeatureID	Type	Length (m)
0040	Bicycle lane	1,500

● node ○ vertex

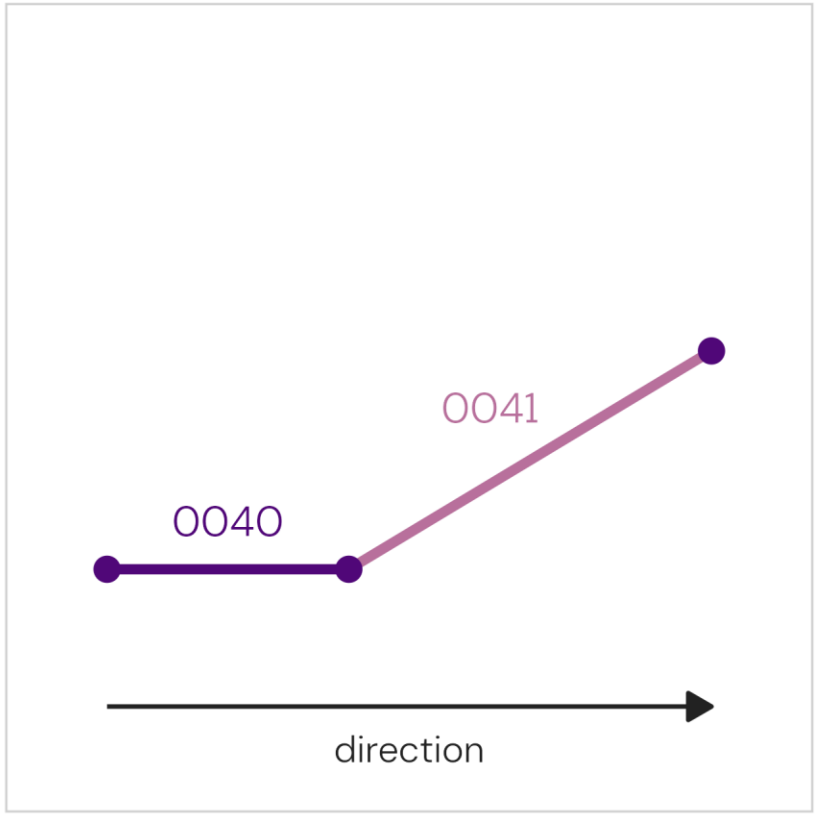
Polyline vector



FeatureID	Type	Length (m)
0040	Bicycle lane	1,650

● node ○ vertex

Polyline vector



FeatureID	Type	Length (m)
0040	Bicycle lane	600
0041	Bicycle lane	1,050

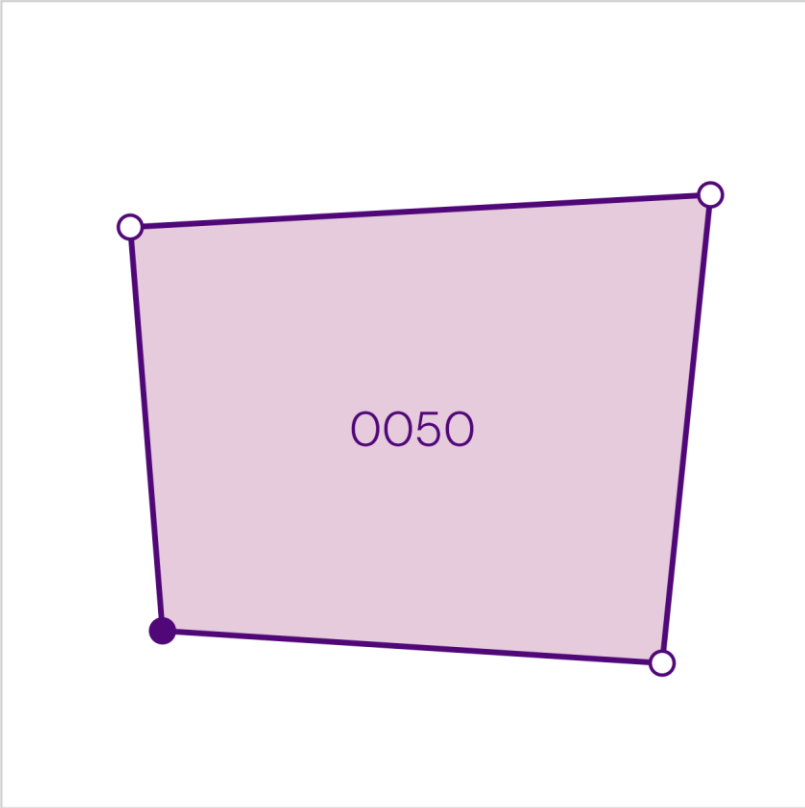
● node ○ vertex

Polygon vector

Characteristics of a polygon vector in a GIS data model:

- Series of XY locations (coordinates) to form an enclosed region.
- Has an area.
- Has a perimeter (the length of its boundary).
- Geometry consists of at least three nodes or vertices whereby the first node or vertex connects with the last one.
- Used for: features with enclosed regions such as buildings and administrative areas.

Polygon vector

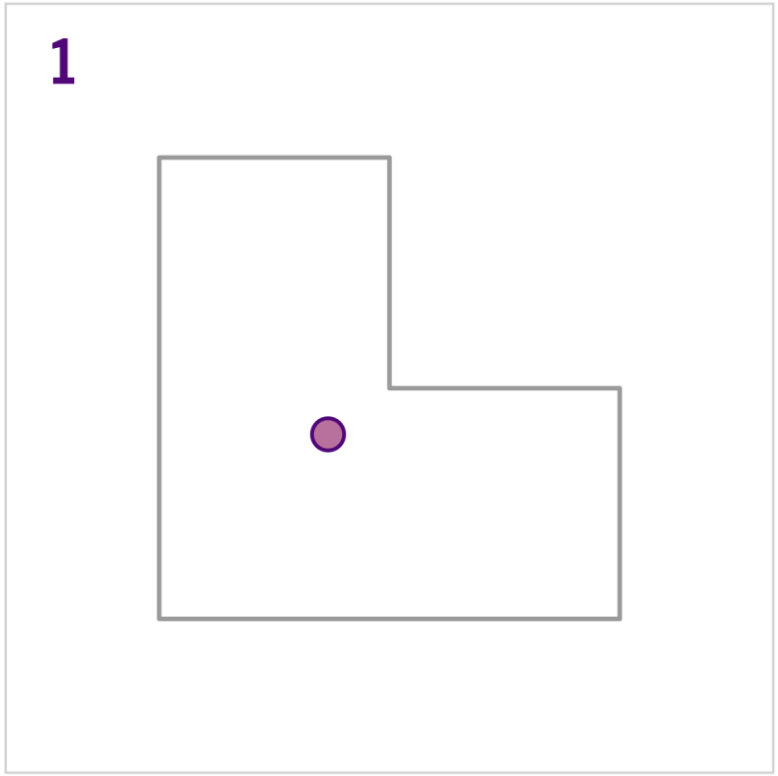


FeatureID	Type	Area (m²)
0050	University building	2000

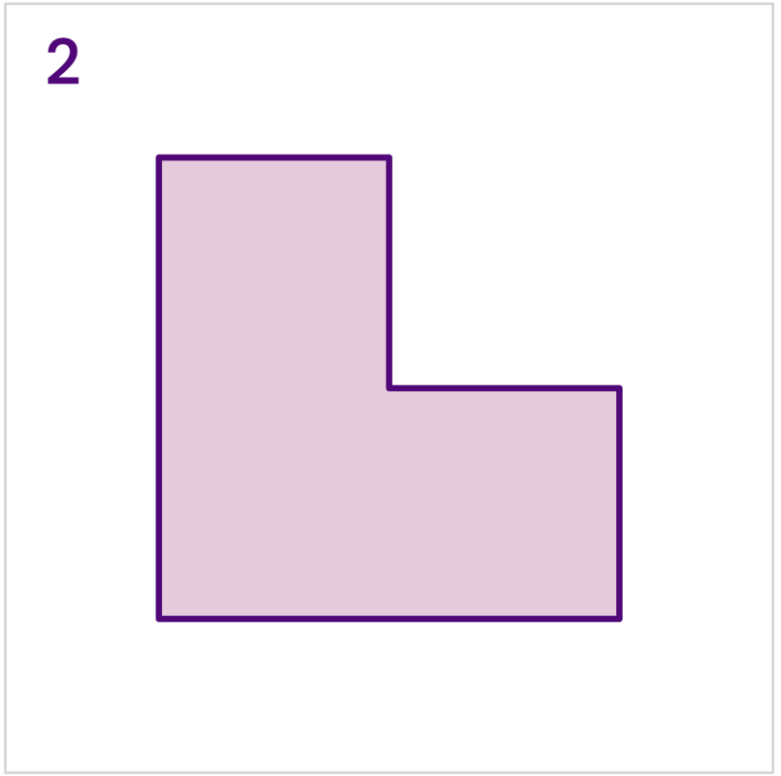
● start and end node ○ vertex

Representation

Representing the North-West Wing Building

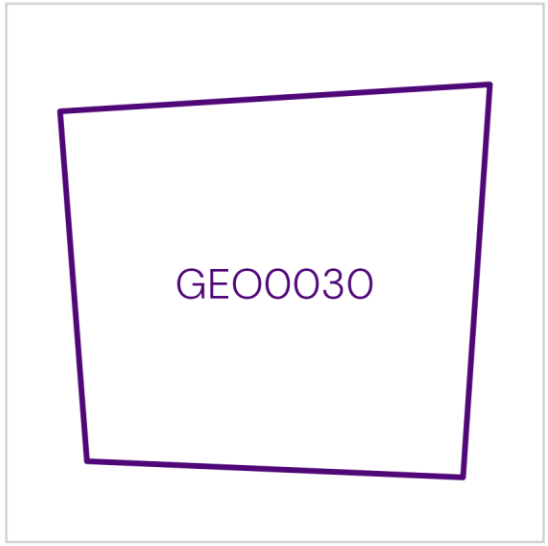


A point



A polygon

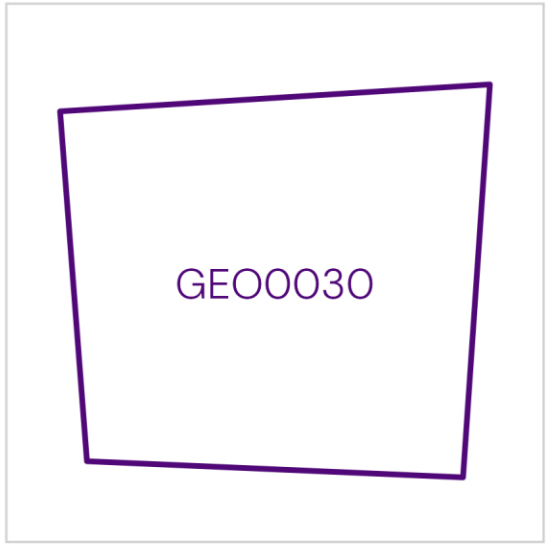
Attribute join: match rows by a shared ID



Features

FeatureID	GeoID
0050	GEO0030

Attribute join: match rows by a shared ID



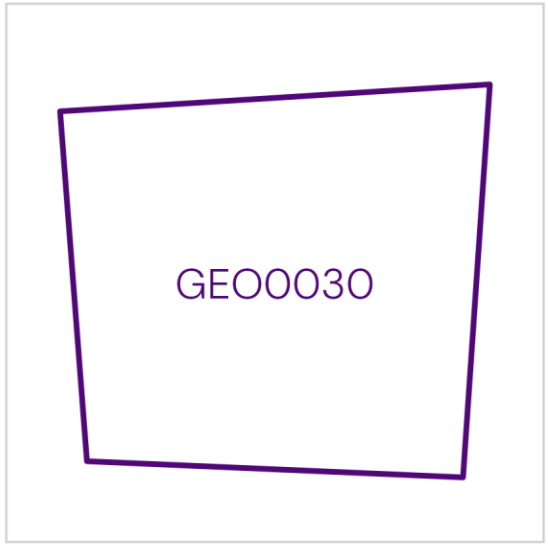
Features

FeatureID	GeoID
0050	GEO0030

Outside data

GeoID	Population
GEO0030	540
GEO0031	320

Attribute join: match rows by a shared ID



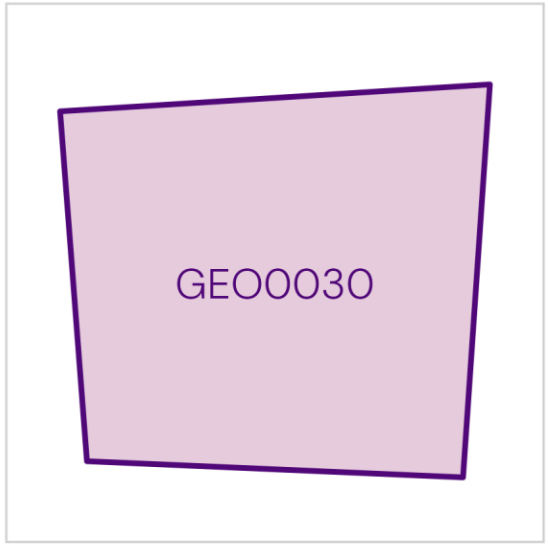
Features

FeatureID	GeoID
0050	GEO0030

Outside data

GeoID	Population
GEO0030	540
GEO0031	320

Attribute join: match rows by a shared ID



Features with the outside data added

FeatureID	GeoID	Population
0050	GEO0030	540

Left join: keep every row of the first table

Table 1

id	a
1	a1
2	a2

Table 2

id	b
1	b1
3	b3
4	b4

Left join

id	a	b
1	a1	b1
2	a2	NA

id 2 has no match in Table 2, so its value from Table 2 is NA.

Invented example: two small tables joined on id.

Inner join: keep only rows found in both

Table 1

id	a
1	a1
2	a2

Table 2

id	b
1	b1
3	b3
4	b4

Inner join

id	a	b
1	a1	b1

Only id 1 appears in both tables, so only that row is kept.

Invented example: two small tables joined on id.

Outer join: keep every row from both tables

Table 1

id	a
1	a1
2	a2

Table 2

id	b
1	b1
3	b3
4	b4

Outer join

id	a	b
1	a1	b1
2	a2	NA
3	NA	b3
4	NA	b4

Every id from either table is kept; a missing match is NA.

Invented example: two small tables joined on id.

Spatial data formats

- Several commonly used file formats that store spatial data exist.
- Different file formats for vector data and raster data.
- Common vector formats: shapefile, GeoJSON, GeoPackage
- Common raster formats: GeoTIFF, GeoPackage

Shapefiles

- Perhaps the most (in)famous file format.
- Widely used, despite being horribly outdated, especially because of the limitations of the `.dbf` format.
- A shapefile is not a single file, but a collection of files of which at least three are needed for the data to be displayed in GIS software.

Shapefiles

- `.shp` contains the feature geometry.
- `.shx` index file which stores the position of the feature's ID in the `.shp` file.
- `.dbf` stores all attribute information associated with the records..
- `.prj` contains the coordinate system information and projection. *Optional*.
- `.xml` general metadata. *Optional*.
- `.cpg` encoding information. *Optional*.
- `.sbn` optimisation file for spatial queries. *Optional*.

Shapefiles



GeoPackage

- A GeoPackage is an open, standards-based, platform-independent, portable, self-describing, compact format for transferring geospatial data.
- It stores spatial data layers in a single file, based upon an SQLite database.
- How to spot in the wild: .gpkg

Spatial data formats

- R has the capacity to read, load and store a range of file formats.
- Functions in both the base R library plus a huge host of software-specific packages (e.g. STATA, SPSS) for reading, writing and converting data between different file formats associated with those specific software (e.g. from a SPSS file to a `csv` etc.).
- Base R does not handle the reading, loading, and storing of spatial data.

Spatial data formats

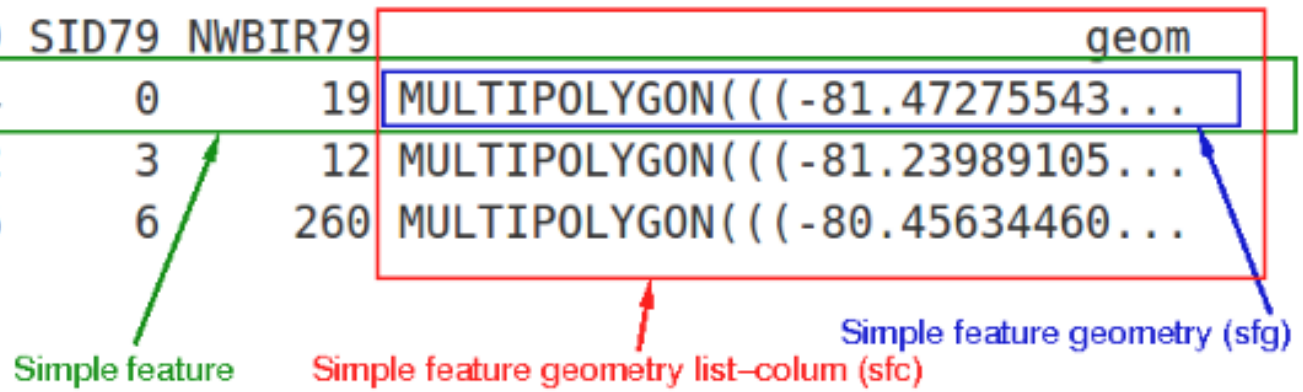
- How do we read in and deal with spatial data?
- GDAL: Geospatial Data Abstraction Library (*reading, writing*)
- GEOS: Geometry Engine Open Source (*spatial operations*)
- PROJ: Cartographic Projection Library (*coordinate transformations)*

GEOS Geometry Engine Open Source

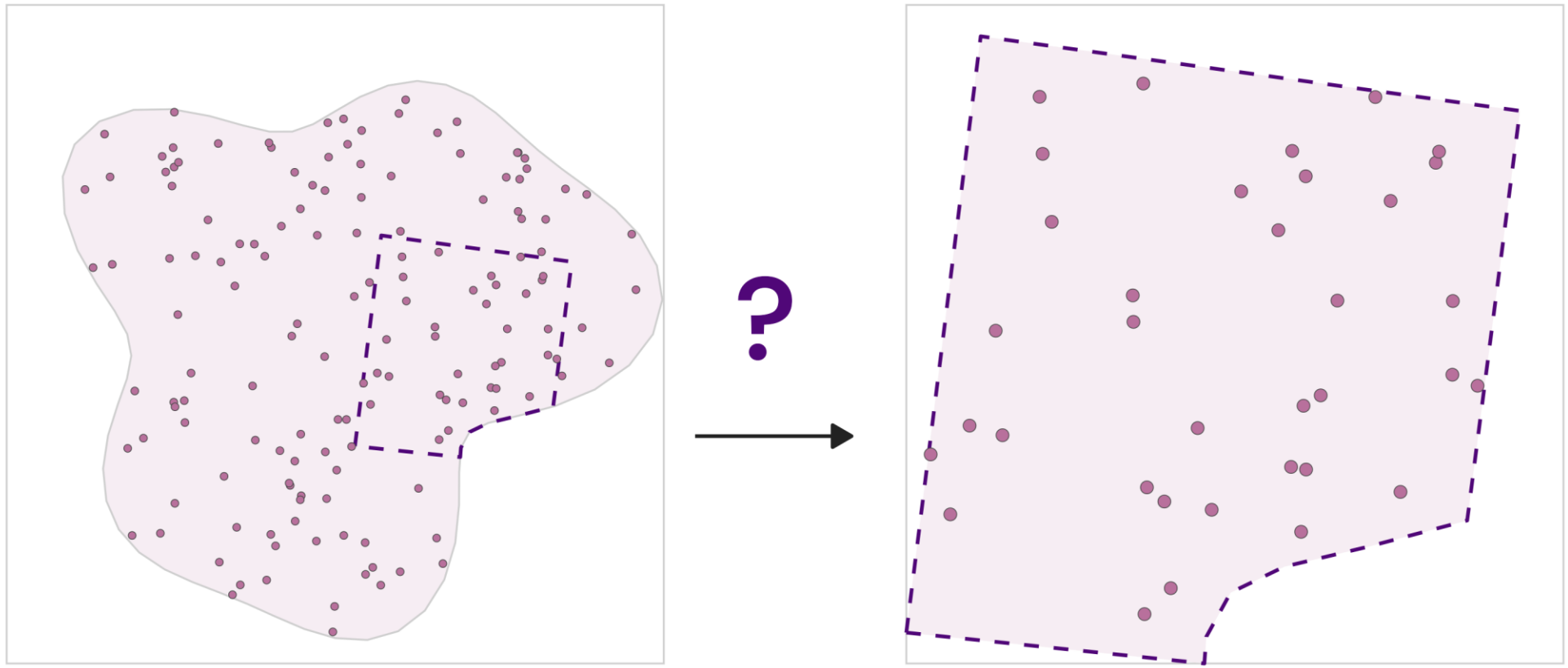


Spatial data formats

```
## Simple feature collection with 100 features and 6 fields
## geometry type:  MULTIPOLYGON
## dimension:      XY
## bbox:           xmin: -84.32385 ymin: 33.88199 xmax: -75.45698 ymax: 36.58965
## epsg (SRID):    4267
## proj4string:    +proj=longlat +datum=NAD27 +no_defs
## precision:      double (default; no precision model)
## First 3 features:
##   BIR74 SID74 NWBIR74 BIR79 SID79 NWBIR79                              geom
## 1  1091     1       10  1364     0       19 MULTIPOLYGON(((-81.47275543...
## 2   487     0       10   542     3       12 MULTIPOLYGON(((-81.23989105...
## 3  3188     5      208  3616     6      260 MULTIPOLYGON(((-80.45634460...
```



Spatial operations

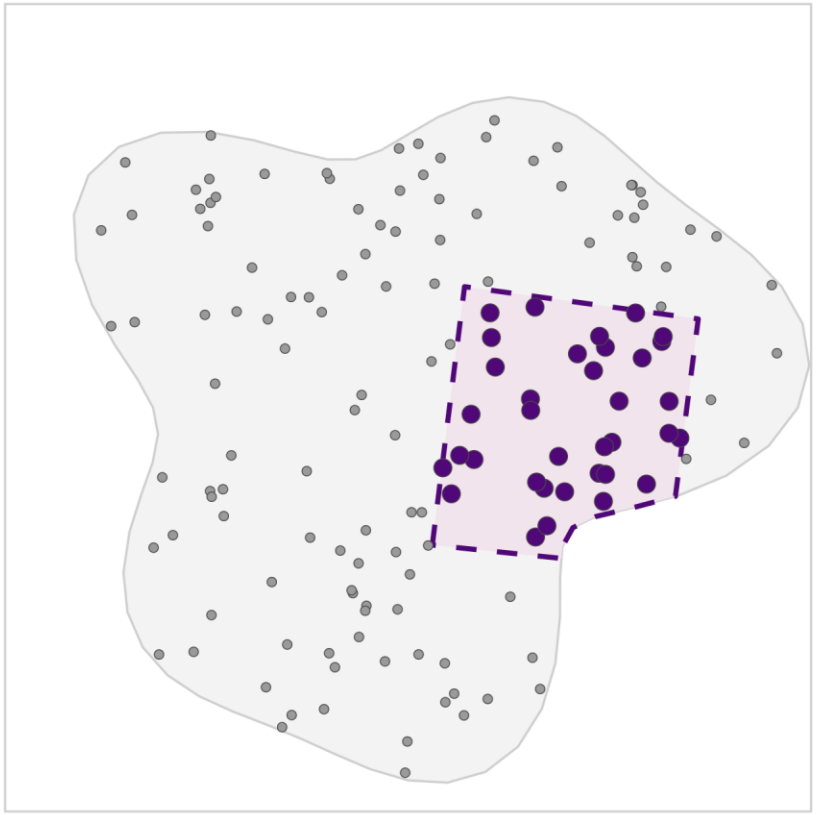


Invented example: 150 points in a region.

Spatial operations

- Spatial relationships describe how features sit in relation to each other, for example within, touching, overlapping or near, often referred to as topological relationships.
- Spatial operations make use of these spatial relationships between spatial objects to combine, identify, extract information.
- A spatial query selects existing features by where they are; a geometric operation creates new shapes.

Spatial query: select features by location

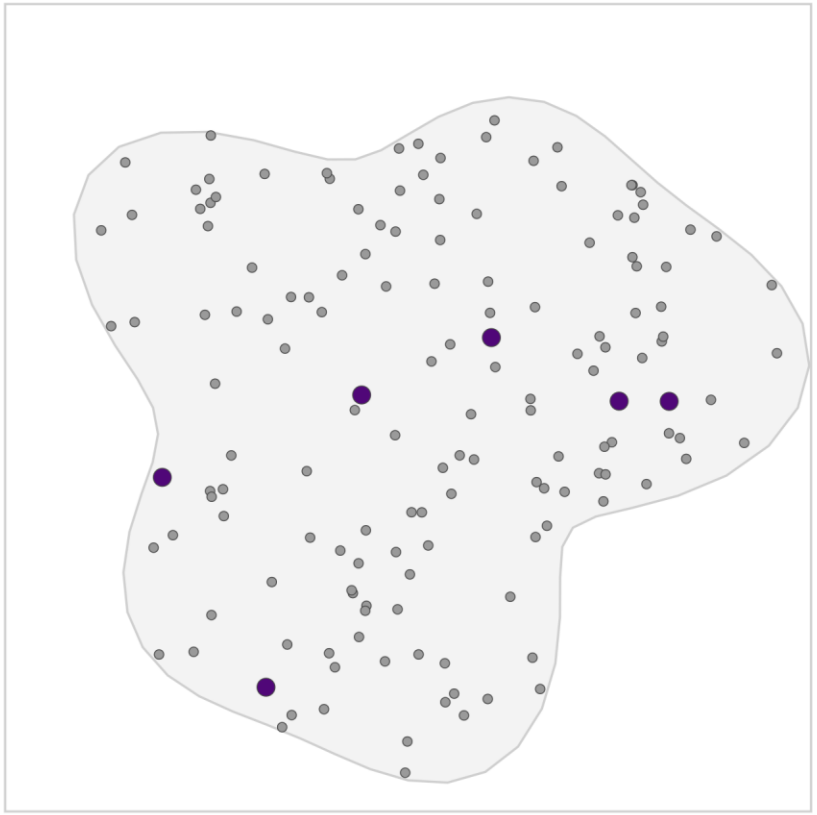


Select by location

- Inside the district: 35
- Outside: 115

Invented example: 150 points in a region.

Attribute query: select features by value

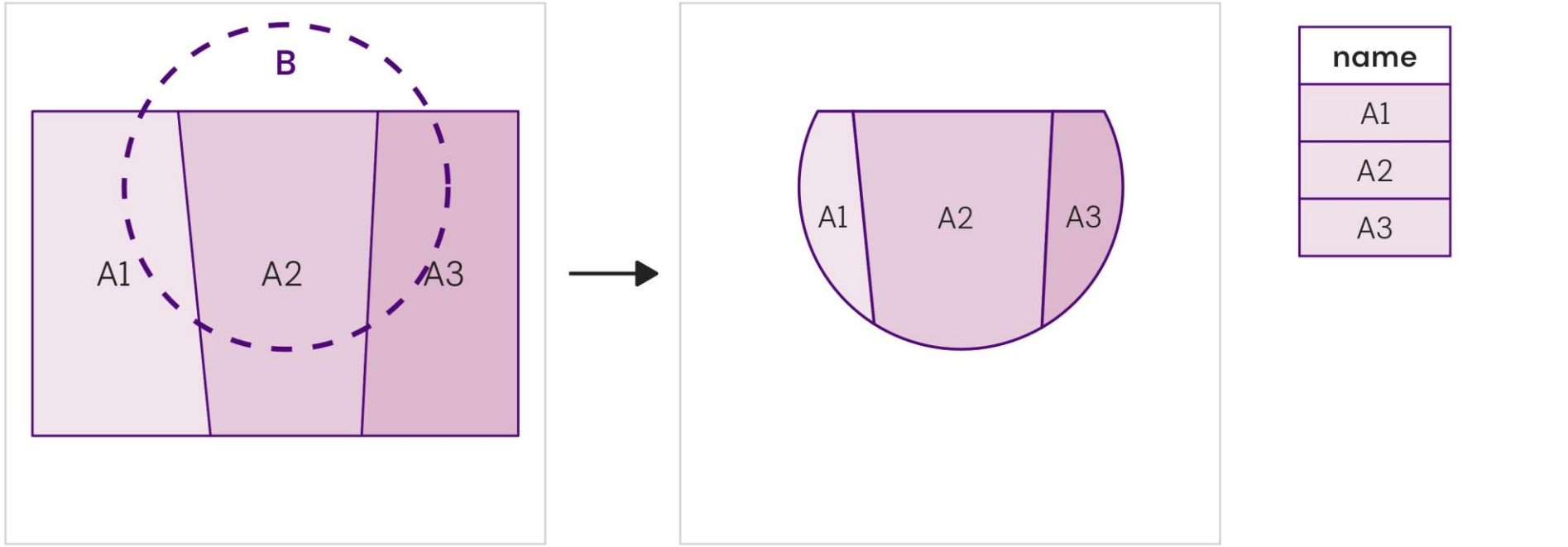


Select by value: above 94

- Selected: 6
- Not selected: 144

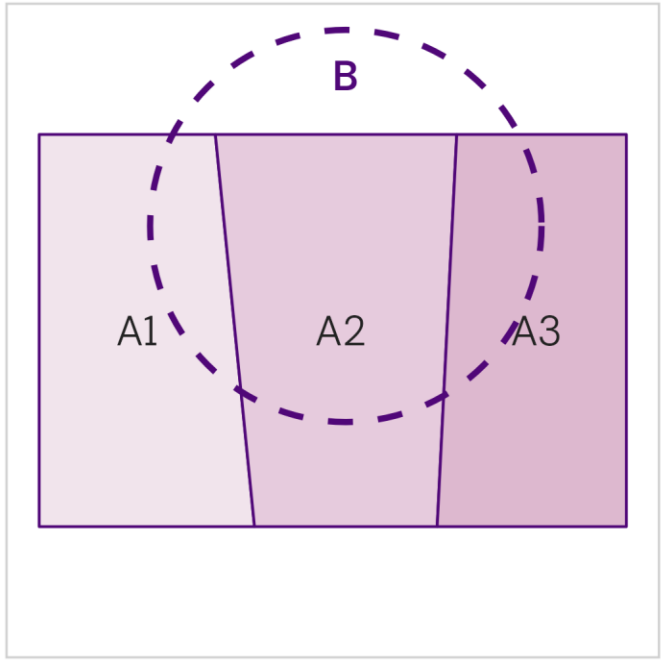
Invented example: 150 points in a region.

Clip: cut one layer to the shape of another

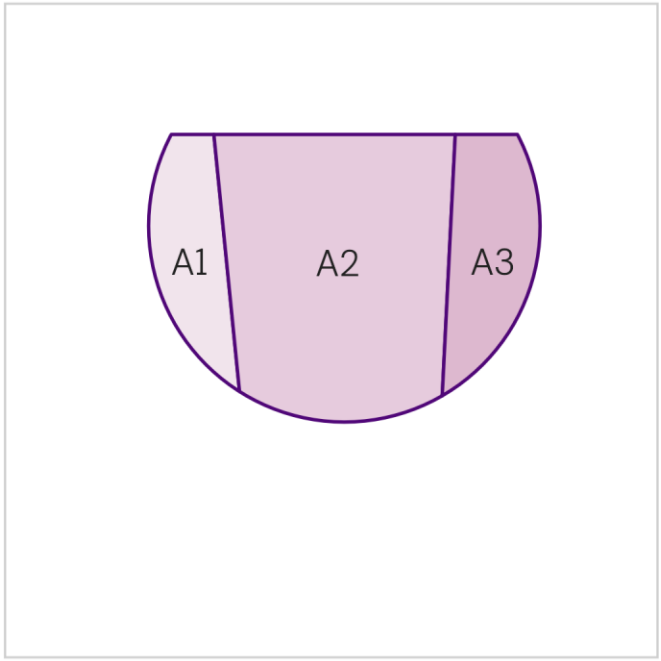


Invented example: three districts (A) and a circle (B).

Intersect: keep only where two layers overlap



Layer A and layer B



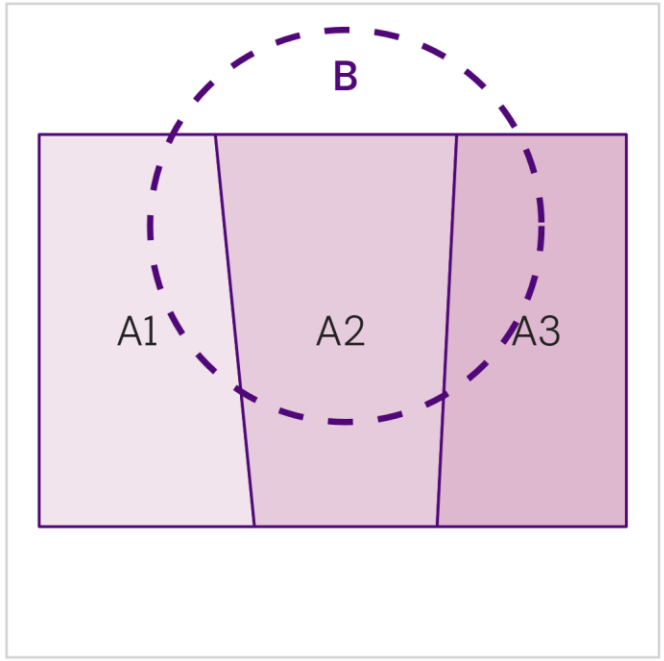
Result

name	zone
A1	B
A2	B
A3	B

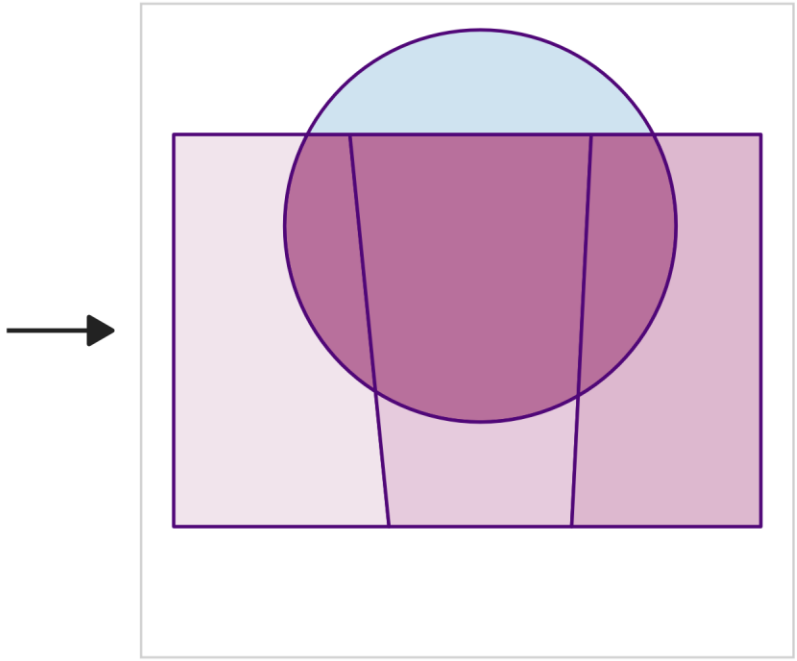
Attribute table

Invented example: three districts (A) and a circle (B).

Union: combine two layers, keep every area



Layer A and layer B



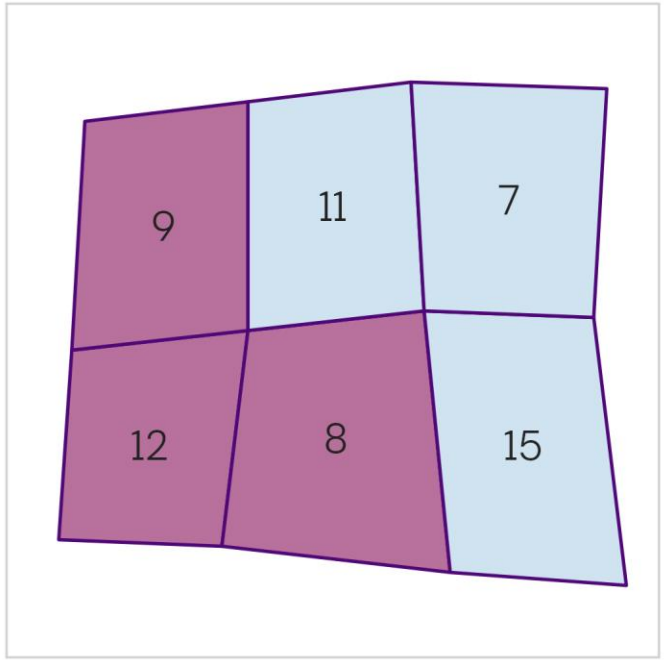
Result

name	zone
A1	B
A2	B
A3	B
A1	NA
A2	NA
A3	NA
NA	B

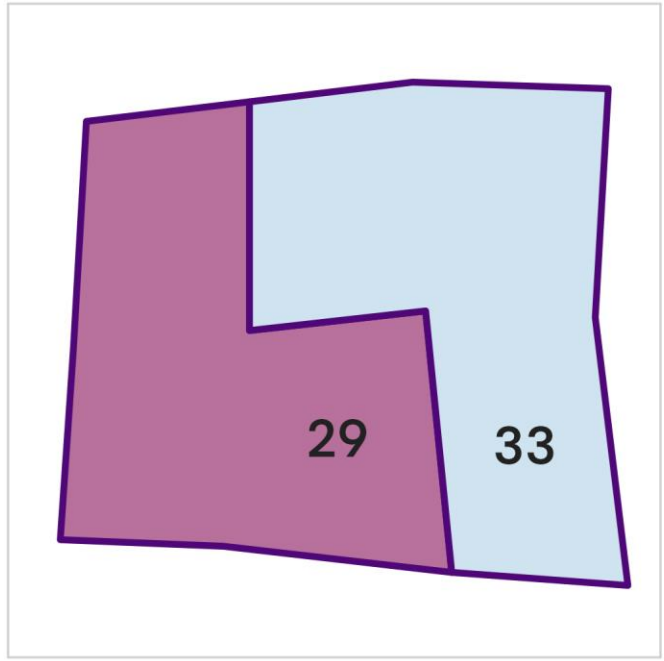
Attribute table

Invented example: three districts (A) and a circle (B).

Dissolve: merge features into one



Six features with a count



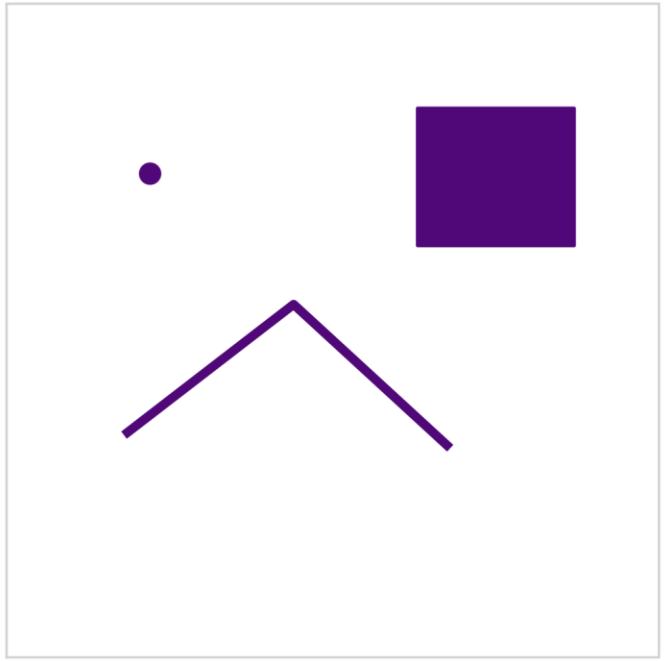
Dissolved by region, counts summed

region	count
X	29
Y	33

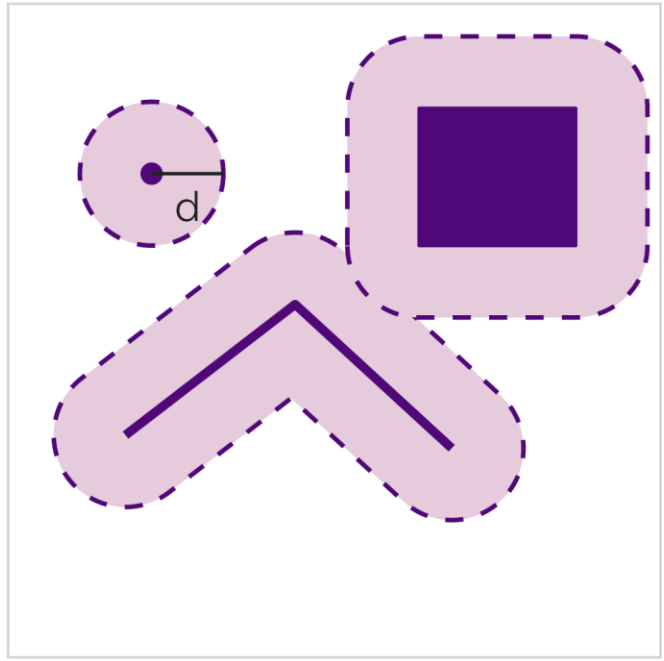
Attribute table

Invented example: six features with a count, dissolved by region.

Buffer: draw a zone of fixed distance



A point, a line and a polygon

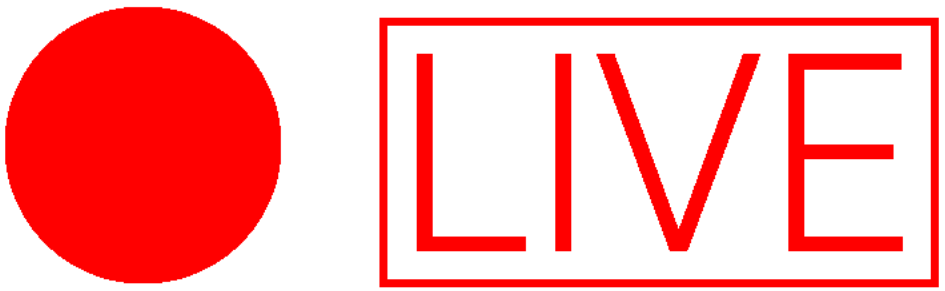


Buffers of distance d

Every buffer reaches the same distance d from its feature, whatever the shape.

Invented example: a point, a line and a polygon.

RStudio



Conclusion

- The core of spatial analysis comes down to conducting spatial queries and executing geometric operations.
- Spatial analysis therefore relies on the spatial properties of an object as well as on the spatial relationships both *within* and *between* spatial objects.

Questions

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